

Assignment 1

Due Date: October 15, 2010 by 5:30 pm

1. (4 points) Given the file format <filename>.<ext>, where <filename > is a file name without a path and with only alphanumeric characters and <ext> is a 3-letter filename extension, write a shell script to strip off the extension of a given filename. Hint: there are generally two ways to strip an extension off of a filename, though one is preferable in this case:

- a. Use the linux utility **basename**
- b. Use the following shortcut (var would store the filename):
`${var%%.*}`

2. (5 points) Copy the following values into a file called energies.dat:

```
1 .00000050 29.538 -17.1677644485
2 .00000346 28.581 -17.1677641803
3 .00000970 26.710 -17.1677637523
4 .00001968 30.000 -17.1677111734
5 .00003150 27.459 -17.1677111652
6 .00004350 25.003 -17.1677112749
7 .00005313 30.000 -17.1676425124
8 .00005952 27.637 -17.1676425364
9 .00006282 25.427 -17.1676425261
10 .00006315 30.000 -17.1675795495
```

Write a shell script to read the lines from a file and write out columns 1 and 4 to the screen. Now, run the script but redirect the output to a file.

3. (6 points) Environment settings – make the following changes to your .bashrc, source it and verify using the linux echo command at the command line:
 - a. Aliases:
 - i. Make a short alias for “ls -allrt”
 - ii. Make a short alias to ssh to one of the ACEnet clusters
 - iii. Make an alias for the rm command to make it ask before deleting a file (hint: check the man page for rm to get this option)
 - b. Paths:
 - i. Add /home/\$USER/bin to your PATH

- ii. Add `/usr/share/man/man1` to your `MANPATH`
4. (5 points) Write a shell script to take some options and for each chosen option, write something to a file.
5. (7 points) Write regular expressions to do the following things:
- Match all uppercase characters in a string
 - Match the numbers 3, 6, and 7 in a numerical sequence
 - Match any word
 - Match one of a few select words in some text
 - Match words that are similar
 - Match the special characters `*`, `?`, `$`, `(`, and `.`
 - Match numbers starting with a digit followed by a period
6. (5 points) Write a perl script to determine the number of words in a line of text read from standard input.
7. (11 points) Write a perl script that reads in Cartesian coordinates for a small set of 3-dimensional grid points from a file, stores it in an associative array, and prints the grid point that was requested as an option to the script.
- To test the script, you can generate a set of labeled grid points (does not matter if they are real) and store them in a file that you can read in with the script.
- Hint: you can use the grid point's label to select the right coordinates from the associative array. The grid point could be coordinates for an atom in a molecule or a point in a mesh used to solve an equation numerically.
8. (9 points) Write a perl script to parse the lines in the text file you created in the first step of question 2. Have it print columns 1, 2, and 3 to a file.
9. (4 points) What do the following regular expressions match?
- `/m.+?e/`
 - `/x + y/`
 - `^d+/`
 - `^S+/`
10. (13 points) Jobs on the ACEnet systems are submitted using a script that begins with a set of directives (lines beginning with a `#`) and follows with shell scripting to run a user's job. Say you have a set of jobs with different input and output files that all follow the same batch script form, given here:

```
#$ -cwd  
#$ -S /bin/bash  
#$ -l h_rt=1:0:0  
#$ -pe $numprocs
```

```
mpirun ./mycode.exe < $infile > $infile.out
```

where the \$infile is defined for each individual job in the set and the batch scripts all have the name \$infile.n1ge. Also, \$numprocs is an option to the script, which will set the number of processors for the run.

Write a perl script that takes a set of input files and creates a set of job scripts using the above format. Assume that the files \$infile and \$infile.n1ge are all in the same directory as the perl script and that there are too many to list them as options to the script.

NOTE: you will need to have files in that directory to use for putting into the n1ge script. However, since it is not likely that you have these to use as a test for the script, for this question you do not need to prove that your script works, just that it is complete.

BONUS: (5 points) assume that the files are not in the same directory as the script used to run them and have your script read them in using absolute path names as part of the filename.